

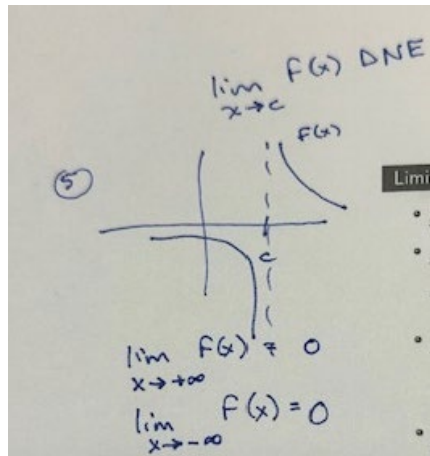
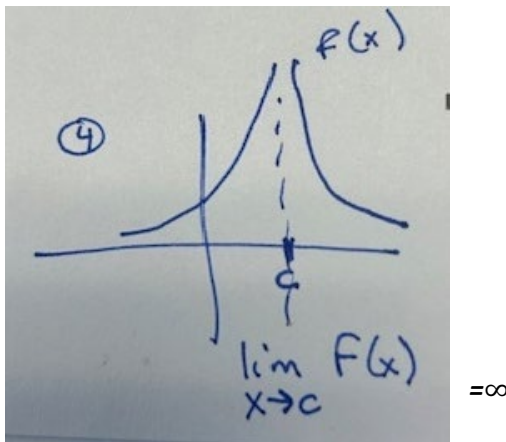
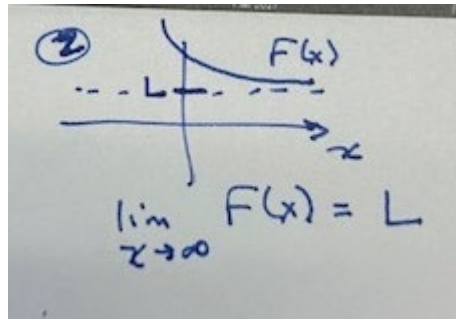
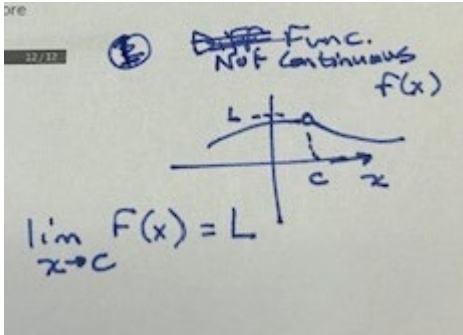
Sec 1.9 – Further Limit Calculations Using Algebra

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

What is the difference between the intervals $(1, 3)$ and $[1, 3]$?

$(1, 3)$ is equivalent to $1 < x < 3$ while $[1, 3]$ is equivalent to $1 \leq x \leq 3$.

For the following, you should be able to produce the graph from the limit definition and the limit definition from the graph.



Partner up

Calculate the following limits:

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2} = 0$$

This function is continuous at $x = 1$ (the denominator is not 0 and both numerator and denominator are polynomials), so we can just substitute $x = 1$ to evaluate the limit.

$$\lim_{x \rightarrow 1} \frac{1}{x-1} \text{ DNE}$$

We cannot just substitute $x = 1$ because it makes the denominator 0, so $\frac{1}{x-1}$ is not continuous at $x = 1$ and we cannot use continuity to evaluate the limit. However, we can think about this: From the right, as x approaches 1, we get a very small positive number in the denominator. As the denominator gets smaller, the overall fraction gets bigger and the limit from the right is ∞ . Similarly, from the left we would get very small negative numbers in the denominator, so the function values are larger and larger negative numbers, so the limit from the left is $-\infty$. Since the limits from the left and right don't match, this limit does not exist.

$$\lim_{x \rightarrow \infty} \frac{2}{x^2} = 0$$

Since ∞ is not a real number, we cannot use continuity. But we think about substituting larger and larger values for x . As the denominator gets larger and larger, the y -values of the function $\frac{2}{x^2}$ gets smaller and smaller, while staying positive. Therefore, the limit goes to 0.

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x - 6}{x + 4} = -\frac{12}{7} \text{ by Continuity}$$

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{(\sqrt{t^2 + 9} - 3)(\sqrt{t^2 + 9} + 3)}{t(\sqrt{t^2 + 9} + 3)} &= \lim_{t \rightarrow 0} \frac{(t^2 + 9) - 9}{t(\sqrt{t^2 + 9} + 3)} \\ &= \lim_{t \rightarrow 0} \frac{t^2}{t(\sqrt{t^2 + 9} + 3)} = \lim_{t \rightarrow 0} \frac{t}{(\sqrt{t^2 + 9} + 3)} = 0 \end{aligned}$$

$$\lim_{x \rightarrow 2} \frac{3x^2 - 4x - 4}{2x^2 - 8} = \lim_{x \rightarrow 2} \frac{(3x + 2)(x - 2)}{2(x - 2)(x + 2)} = \lim_{x \rightarrow 2} \frac{3x + 2}{2(x + 2)} = \frac{8}{8} = 1$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3}}{x} &= \lim_{x \rightarrow 0} \frac{(\sqrt{3+x} - \sqrt{3})(\sqrt{3+x} + \sqrt{3})}{x(\sqrt{3+x} + \sqrt{3})} \\ &= \lim_{x \rightarrow 0} \frac{(3+x) - 3}{x(\sqrt{3+x} + \sqrt{3})} = \lim_{x \rightarrow 0} \frac{1}{(\sqrt{3+x} + \sqrt{3})} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6} \\ &= \lim_{h \rightarrow 0} \frac{h(9h + 12)}{h} = \lim_{h \rightarrow 0} 9h + 12 = 12 \end{aligned}$$

We divide both numerator and denominator by the highest power of x in the denominator to make the limit of the denominator not 0. (This is multiplying by a “fancy 1.”)

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{3x^2 + 2x + 5}{2x^2 - 3x} &= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} 3x^2 + 2x + 5}{\frac{1}{x^2} 2x^2 - 3x} \\ &= \lim_{x \rightarrow \infty} \frac{3 + \frac{2}{x} + \frac{5}{x^2}}{2 - \frac{3}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} (3 + \frac{2}{x} + \frac{5}{x^2})}{\lim_{x \rightarrow \infty} (2 - \frac{3}{x^2})} \quad \text{by the Quotient law} \\ &= \frac{3}{2}\end{aligned}$$

$$\lim_{x \rightarrow \infty} 4 + \frac{5}{x} - \frac{3}{x^2} = \lim_{x \rightarrow \infty} 4 + \lim_{x \rightarrow \infty} \frac{5}{x} - \lim_{x \rightarrow \infty} \frac{3}{x^2} = 4 + 0 + 0 = 4$$

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{2x^4 - 3x^2 + 4}{7x^4 + 2x} &= \lim_{x \rightarrow -\infty} \frac{2x^4 - 3x^2 + 4}{7x^4 + 2x} \cdot \frac{\frac{1}{x^4}}{\frac{1}{x^4}} \\ &= \lim_{x \rightarrow -\infty} \frac{2 - \frac{3}{x^2} + \frac{4}{x^4}}{7 + \frac{2}{x^3}} = \frac{\lim_{x \rightarrow -\infty} (2 - \frac{3}{x^2} + \frac{4}{x^4})}{\lim_{x \rightarrow -\infty} (7 + \frac{2}{x^3})} = \frac{2}{7}\end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{3x^2 - 7}{4x^3 + 5x^2 - 3x} &= \lim_{x \rightarrow -\infty} \frac{3x^2 - 7}{4x^3 + 5x^2 - 3x} \cdot \frac{\frac{1}{x^3}}{\frac{1}{x^3}} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{3}{x} - \frac{7}{x^3}}{4 + \frac{5}{x} - \frac{3}{x^2}} = \frac{\lim_{x \rightarrow -\infty} (\frac{3}{x} - \frac{7}{x^3})}{\lim_{x \rightarrow -\infty} (4 + \frac{5}{x} - \frac{3}{x^2})} = \frac{0}{4} = 0\end{aligned}$$

$$\lim_{x \rightarrow -\infty} \frac{3x^3 + 2x + 5}{2x^2 - 3x} = \lim_{x \rightarrow -\infty} \frac{3x + \frac{2}{x} + \frac{5}{x^2}}{2 - \frac{3}{x^2}} = -\infty$$

We can't use the sum law to finish this since $\lim_{x \rightarrow -\infty} 3x = -\infty$ and DNE. BUT, we can reason that the numerator is getting more and more negative while the denominator has a limit of 2 and therefore the overall fraction keeps getting more and more negative.